

Approach to Postural Orthostatic Tachycardia Syndrome

Nathaniel M. Robbins,^{1,2} Elisabeth P. Golden,³ Kelly C. Freeman,⁴ Ratna K. Bhavaraju-Sanka,⁵ Howard Snapper,⁶ and Glen A. Cook⁷

Neurol Clin Pract. 2026;16:e200620. doi:10.1212/CPJ.0000000000200620

Correspondence

Dr. Robbins
NRobbins3@mgh.harvard.edu

Abstract

Purpose of Review

To support general neurologists, advanced practice providers, and general practitioners in the diagnosis and treatment of postural tachycardia syndrome (POTS) and related dysautonomias. This review and approach are based on the authors' clinical experience, nonsystematic literature review, and expert opinion, with experts drawn from the American Autonomic Society Education Committee.

Recent Findings

POTS is a common condition seen in general practice. The diagnosis can be easily made in a nonspecialist setting when patients meet the syndromic criteria for POTS: (1) chronic orthostatic intolerance (bothersome symptoms brought on by upright posture, and relieved by lying down), (2) excessive orthostatic tachycardia (heart rate augmentation with standing greater than 30 beats per minute in adults), (3) absence of orthostatic hypotension, and (4) reasonable exclusion of other causes (e.g., dehydration, anemia, hyperthyroidism). There are many causes, but the unifying perturbations are low effective circulating volume with decreased cerebral blood flow and a hyper-sympathetic state. Education and nonpharmacologic therapies such as salt, water, and exercise are first-line treatments. Several safe and effective medication options are also available, such as propranolol and midodrine. Several comorbidities are common, such as migraine and neurocardiogenic syncope, and should be screened for and treated rather than attributed out-of-hand to POTS.

Summary

POTS and related conditions are common. Initial diagnostic and treatment steps can be undertaken in a general practice setting. Referral to specialists is appropriate for refractory cases or when the diagnosis is uncertain.

Illustrative Case

A 24-year-old woman presents with frequent presyncope and recurrent syncope. She is in physical therapy school and fainted last week when she had to stand for quite a long time during hospital rounds. She has noticed lightheadedness and a racing heart upon standing for many years, but she always thought it was normal. These symptoms limited her ability to play running sports, though she was a competitive swimmer in high school. She reports shortness of breath, tunnel vision, presyncope, and palpitations whenever she stands for a long time. It is better walking than standing still. It is worse in hot weather—for the last year she has been sitting down during showers. She has an Apple watch, and she noticed her heart rate goes up to 150 beats per minute often with little provocation—just standing or walking up stairs can do it. She feels better sitting down, and now spends most of the day sitting or lying down.

Background

POTS is a syndrome defined by consensus criteria, based on the presence of chronic orthostatic intolerance and symptomatic orthostatic tachycardia without hypotension (Table 1).¹⁻⁴

¹Department of Neurology, Mass General Brigham, Boston, MA; ²Department of Neurology, Harvard Medical School, Boston, MA; ³Department of Medicine, Neurology Section, UT Health Science Center at Tyler, Tyler, TX; ⁴The Dysautonomia Project, Clearwater, FL; ⁵Department of Neurology, University of Texas Health San Antonio, TX; ⁶Department of Cardiology, Wellstar Healthcare System, Marietta, GA; and ⁷Department of Neurology, Uniformed Services University, Bethesda, MD.

Bothersome symptoms such as forceful tachycardia or presyncope should be relieved substantially by lying down but may not fully resolve. POTS can cause significant disability and impaired quality of life.^{5,6} It has an estimated worldwide prevalence of 0.2%–1%, with a predilection for females age 15–45.¹

POTS can be diagnosed based on history and orthostatic vital signs. When diagnosing POTS, alternative causative medical conditions should be reasonably excluded (see below), but specialized testing is not necessary unless additional factors call for it. Patients with POTS may not meet heart rate (HR) criteria every time they are tested, as symptoms and HR tend to improve in the afternoon with diurnal variability⁷ and fluctuate day-to-day. Patients who do not meet HR criteria can be characterized as having orthostatic intolerance, or “Postural Symptoms Without Tachycardia,”⁴ which sometimes responds similarly to POTS. Evaluation and management of POTS can be undertaken in the primary care or specialist setting, with autonomic subspecialty referrals reserved for complicated or refractory cases, or when the diagnosis is not certain.⁸

Although POTS is a heterogeneous syndrome, the common unifying physiologic and clinical features are manifestations of decreased central blood volume (i.e., blood in the heart, lungs, and great vessels) in the upright position, often with accompanying symptoms of a hypersympathetic state.⁹ Although the diagnostic features of POTS are defined by the orthostatic symptoms, other symptoms are often present (Table 2).

“NOTS”: Conditions Often Incorrectly Mislabeled as POTS

Many other conditions are incorrectly called “POTS.” For example, vasovagal and neurally mediated syncope are characterized by paroxysmal attacks of decreased cerebral perfusion. These can coexist in POTS but are distinct. A patient can have POTS without syncope, syncope without POTS, or both together.

Similarly, many patients have paroxysmal attacks of tachycardia at rest, because of anxiety or panic, pain, supraventricular tachycardia, pheochromocytoma, pseudo-pheochromocytoma, or unknown etiology. Such paroxysmal episodes can occur in patients without POTS or can co-occur with POTS but should not be confused with POTS. Some patients can have

Table 2 Common Symptoms of POTS or Its Comorbidities

Neurologic	Dizziness or lightheadedness Cognitive dysfunction/brain fog Paresthesia Generalized weakness Headache, especially migraines Fatigue Nonrestorative sleep
Cardiovascular	Orthostatic intolerance Postural tachycardia Syncope and presyncope Chest discomfort Dependent venous stasis
Gastrointestinal (GI)	Nausea/vomiting Abdominal pain Early satiety Bloating Diarrhea or constipation Multiple food intolerances Reflux
Respiratory	Shortness of breath Hyperventilation
Thermoregulatory	Heat or cold intolerance Hyperhidrosis
Genitourinary	Urinary frequency, urgency Pelvic pain
Pupillomotor	Blurred vision Difficulty with bright/flashing lights
Skin/musculoskeletal	Multiple allergies Frequent hives Flushing Joint hypermobility Joint pain

Abbreviation: POTS = postural tachycardia syndrome.

tachycardia with exercise, which is also not POTS. POTS is a disorder of chronic orthostatic intolerance with symptoms relieved by lying down and should not be confused with other causes of paroxysmal sympathetic hyperactivity (often described by patients as “adrenaline surges”).

Evaluation

History

The history of present illness (HPI) is the most informative and important part of the evaluation of POTS.¹⁰ History taking should start with an open-ended question, such as

Table 1 Diagnostic Criteria for POTS

Frequent symptoms (e.g., lightheadedness, palpitations, tremor, generalized weakness, presyncope, fatigue) that occur with standing are relieved substantially by recumbency and persist for longer than 3 mo
Sustained increase in heart rate of ≥30 beats per minute (bpm) when moving from a recumbent to a standing position (or ≥40 bpm in individuals 12–19 y of age)
The absence of orthostatic hypotension (≥20 mm Hg drop in systolic blood pressure or ≥10 mm Hg in diastolic blood pressure)

Abbreviation: POTS = postural tachycardia syndrome.

“What is bothering you today?” Orthostatic intolerance is typically the core clinical complaint, manifesting as racing heart, lightheadedness/presyncope, dyspnea, palpitations, and other symptoms brought on by upright position. Symptoms are often worse standing still and better when activated muscles promote circulation. Once the core features are elucidated, it is important to flesh out the timeline with questions such as, “When was the last time you felt healthy?” or, “What was the first thing that went wrong?” Many patients with POTS have symptom onset as a teenager or young adult. Subacute onset is also common and reported in autoimmune conditions such as Sjogren disease or after a period of illness, inactivity, or viral infection (such as post-COVID POTS).

The core symptoms of POTS are related to low central blood volume. Decreased venous return to the heart results in decreased stroke volume and is accompanied by sympathetic activation and increase heart rate, presumably to maintain cardiac output and perfusion. Symptoms result from either low cerebral perfusion (e.g., lightheadedness/presyncope) or excess sympathetic activity (e.g., tachycardia, palpitations, jitteriness). Hyperpnea is also seen, as are other symptoms such as brain fog and paresthesia. Many other symptoms are common and can arise as part of POTS or common comorbidities (Table 2).

Patients with POTS often worsen with factors that lower cerebral perfusion and systemic blood pressure (BP), such as elevated temperature (because of heat-induced cutaneous vasodilation), volume depletion, vasodilatory substances (such as alcohol or medications), large meals (when blood is shunted to the splanchnic circulation and vasoactive gut peptides are released), and prolonged standing (with gravitational blood pooling). A review of systems (ROS) should cover autonomic symptoms such as sudomotor function (“Do you sweat normally?”), urinary function, gastrointestinal function, sicca symptoms, and sexual function. A thorough medication history should also be taken, as many common medications can cause or worsen symptoms of orthostatic intolerance or autonomic dysfunction (Table 3).

Associated Syndromes

Several other systemic syndromes are often seen in patients with POTS. These include hypermobile Ehlers-Danlos syndrome or hypermobility spectrum disorder, limited or partial autonomic neuropathies, chronic fatigue syndrome, migraine, functional gastrointestinal disorders, and features typically associated with excessive mast cell activation. Symptoms of these disorders (Table 2) should also be queried.

Depressive symptoms may be present, although estimates of prevalence vary widely.¹¹ Patients with POTS are commonly perceived as “anxious.” However, anxiety is a subjective sensation, and anxiety should be distinguished from the physical symptoms that accompany sympathetic hyperactivity. Some patients with POTS can also report symptoms of fatigue,

cognitive dysfunction, or “brain fog,” unrelated to posture. The nature and pathogenesis of this remains incompletely understood, and comorbidities may be involved.

Objective Evaluation of POTS

Clinical history is the most important component of the medical evaluation in POTS. Because symptoms of POTS and the common comorbidities span organ systems, great care must be taken to determine which symptoms need further work-up and which can be safely attributed to POTS. For example, chest pain and exercise intolerance are common in POTS, but it remains imperative to rule out dangerous conditions (e.g., angina, pulmonary embolism) based on the clinical scenario.

The basic workup of POTS includes a history and physical examination, orthostatic vital signs (detailed below), and a 12-lead electrocardiogram.³ Additional studies to consider are summarized in Table 4.

Orthostatic vital signs are best measured with an active stand test. BP and HR should be measured in a supine patient after the patient rests quietly for at least 5–10 minutes, and then again periodically after immediately standing (without sitting in between) for at least 5 minutes, and ideally for 10 minutes.⁴ Some patients who do not meet the threshold for excessive orthostatic tachycardia with active standing could have greater HR augmentation with tilt table testing.¹²

If patients endorse symptoms of other common comorbidities, such as irritable bowel, chronic fatigue, pain, or headaches, these should also be addressed separately and treated similarly to patients with these conditions who do not have POTS, rather than attributing them to the POTS itself. A full discussion is not in the scope of this paper; several recent review articles provide a more detailed look at these conditions.^{13–17} It can be helpful to ask patients to quantify what percent of their morbidity is because of orthostatic intolerance and to focus care on the most important and bothersome conditions. There remains uncertainty about which of these common comorbidities, if any, are mechanistically related to POTS.

Treatment

Symptomatic treatments for POTS start with a foundation of education and nonpharmacologic interventions—see Table 5. Medications are added as indicated for control of specific symptoms. Education is a key component of the treatment of POTS. Discussing the syndromic nature of the diagnosis, underlying pathophysiology, prognosis (not life-threatening, although at times disabling), and treatment options empowers patients to manage their symptoms more effectively and confidently. Careful attention should be paid to medicines that exacerbate POTS (Table 3). Nonpharmacologic measures aim to increase blood volume and promote venous return to the heart.¹⁸ These include

Table 3 Medications That May Cause or Exacerbate Orthostatic Tachycardia

Medication or class	Mechanism/notes
Diuretics	Volume depletion
Certain antihypertensives	Vasodilation
Anticholinergics	Exacerbate tachycardia
Tricyclic antidepressants	Alpha-1 antagonism causing orthostatic hypotension and anticholinergic effects causing tachycardia
Serotonin-norepinephrine (NE) or NE-selective reuptake inhibitors	Increased norepinephrine can worsen hyperadrenergic symptoms
Amphetamine derivatives or bupropion	Sympathomimetic effect can worsen hyperadrenergic symptoms

increasing fluid and salt intake and physical measures to promote venous return.

Physical exercise is a cornerstone of treatment. Appropriate exercise can reverse some of the pathophysiologic mechanisms underlying the orthostatic intolerance and tachycardia and improve quality of life.²⁶ Exercise for POTS is generally based on a progressive conditioning regimen. Such graded programs start with recumbent exercise (such as recumbent bicycling, rowing, or swimming) to avoid the increased stress of gravity, on top of exercise, and over time increase the duration, intensity, and amount of upright exercise. Resistance and core training should also be performed. Exercises may need to be modified for those with hypermobility to avoid joint pain. Various effective protocols are published.^{9,23,27} There is controversy about how best to rehabilitate patients with significant postexertional malaise; recent work has focused on addressing such barriers to exercise into an effective rehabilitation program.²⁴ Avoiding prolonged bedrest is crucial to prevent superimposed secondary deconditioning.

Pharmacologic interventions are available (all off-label) to mitigate many of the common symptoms of POTS. These

medications increase blood volume, promote vasoconstriction, lower heart rate, or otherwise alter sympathetic signaling (Table 6). Medications for POTS are not “one size fits all” but rather should be carefully chosen to address the most bothersome symptoms. Other medical conditions should be balanced, because some medications may serve dual purposes, and many can exacerbate comorbidities. Pharmacotherapy is discussed more thoroughly elsewhere.^{28,29} Unfortunately, many patients with POTS go for many years without getting prescription treatment, although many treatments are evidence-based, effective, and safe, and appropriate to start in a primary care setting. Low-dose propranolol is effective for exercise intolerance and hyperadrenergic symptoms, whereas pressors such as midodrine are particularly effective in individuals with baseline low blood pressure. Both midodrine and fludrocortisone help prevent recurrent vasovagal neurocardiogenic syncope.^{30,31}

Illustrative Case Resolution

Orthostatic vital signs in the office confirmed excessive HR augmentation without orthostatic hypotension. ECG, thyroid tests, and routine laboratory results were unremarkable.

Table 4 Recommended Workup for Patients With Suspected POTS

Minimum workup	Orthostatic vital signs (see text for details), 12-lead electrocardiogram, complete blood count, chemistry panel, thyroid function tests
Tests to strongly consider	Antinuclear antibodies, Sjogren antibodies, ambulatory cardiac rhythm monitoring, iron studies
Studies to consider in reasonable clinical scenarios	Morning cortisol, echocardiogram, exercise stress test, B12, serum protein electrophoresis, diabetes screening, tryptase or other mast cell activation markers, inflammatory markers, celiac antibodies, serum or urine osmolality, antiphospholipid antibodies, other autoantibodies
Unclear diagnosis or refractory symptoms	Refer to specialist

Abbreviation: POTS = postural tachycardia syndrome.

Table 5 Nonpharmacologic Measures for Management of POTS

Intervention	Details	Purpose	Notes
Increased fluid intake ¹⁹	>2–3 L per day	Volume expansion	Can be plain water or may include electrolyte beverages. Avoid sports drinks with high carbohydrate load and relatively little sodium
Increased sodium ²⁰	~4–6 g sodium per day (>8–12 g salt)	Volume expansion	If no other contraindication (e.g., kidney or heart failure). Salt tablets can be poorly absorbed or cause GI upset)
Compression garments ²¹	Thigh or waist high compression stockings, compression shorts, or abdominal binder	Increased venous return	Stockings must be at least thigh high to be effective. Abdominal binder and shorts are typically easier to use and effective, with better compliance
Physical countermeasures (for acute presyncope) ²²	Several described	Acute mitigation of orthostatic symptoms or presyncope	Leg crossing with thigh and gluteal clenching commonly used
Physical exercise ^{9,18,23–25}	Recumbent exercise, progressive increase in intensity, duration, and transition to upright exercise over time	Increases blood volume and cardiac mass. Provides epinephrine exposure	Resistance and core exercises must be included as well. Published protocols available

Abbreviation: POTS = postural tachycardia syndrome.

A diagnosis of POTS was made. The patient was educated about increasing salt and water intake and started on a graded exercise program. This improved, but she was still symptomatic. Midodrine 5 mg each morning was added, because

symptoms were worse in the morning, and this allowed her to shower normally. A midday dose was added during the summer month. She often stops medications all together during the winter.

Table 6 First-Line Pharmacologic Treatments for Management of POTS

Medication	Typical dosing	Indication	Notes
Midodrine ^{32,33}	2.5–10 mg, morning and mid-day	Hypotension, near syncope and syncope, venous pooling	Afternoon dose may be added. Avoid “round the clock” dosing because of tachyphylaxis. Effect lasts ~3–4 h. Safe
Propranolol ³⁴ (or other nonselective beta-blockers with CNS penetration)	10–20 mg daily to TID	Palpitations, anxiousness, tachycardia, hypertension	Caution in patients with asthma, low blood pressure, or history of anaphylaxis requiring epinephrine. Safe
Pyridostigmine ²⁹	30–60 mg TID	GI symptoms including bloating, constipation; may be useful for tachycardia and other symptoms	Helpful for constipation. May cause increased bowel movements, diarrhea, muscle or abdominal cramping, or excessive mucous production from procholinergic effects. Safe
Fludrocortisone ^{35,36}	0.1–0.2 mg daily	Low volume–related symptoms (tachycardia, dizziness), hypotension, near syncope and syncope	Caution regarding hypokalemia. Potassium supplementation (10 meq KCl per 0.1 mg fludrocortisone) may prevent tachyphylaxis. May cause or exacerbate headaches, or rarely cause mild corticosteroid effects
Ivabradine ³⁷	5 mg BID	Palpitations related to tachycardia	May increase to 7.5 mg BID. Photopsias common. Contraception required for women of childbearing potential
Guanfacine ³⁸	1 mg QHS	Tachycardia, anxiousness, hypertension	May lower blood pressure
Intravenous normal saline ¹⁹	1–2 L as needed	Low volume related symptoms (tachycardia, dizziness), hypotension, near syncope and syncope	Can be used as “rescue” treatment but should be avoided chronically because of risk of complications of long-term IV access. Oral hydration is preferred when possible

Abbreviation: POTS = postural tachycardia syndrome.

Conclusion

POTS is a common condition characterized by chronic orthostatic intolerance, with symptoms brought on by standing and relieved by recumbency, accompanied by excessive heart rate increase upon standing without orthostatic hypotension. General neurologists, primary care providers, general cardiologists, and other nonspecialists can easily make the diagnosis in patients who fit the syndromic criteria. Education and nonpharmacologic measures are first-line treatment, although there are several safe and effective first-line treatments that can be added to help symptoms and quality of life. Patients without a clear diagnosis, or who are refractory to treatment, warrant subspecialty evaluation.

Author Contributions

N.M. Robbins: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. E.P. Golden: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. K.C. Freeman: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. R.K. Bhavaraju-Sanka: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. H. Snapper: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. G.A. Cook: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

Study Funding

This work was supported by the National Institutes of Health. This manuscript is the result of funding in whole or in part by the National Institutes of Health (NIH). It is subject to the NIH Public Access Policy. Through acceptance of this federal funding, NIH has been given a right to make this manuscript publicly available in PubMed Central on the Official Date of Publication, as defined by NIH.

Disclosure

N. Robbins received research support from the National Institute of Health and Theravance Biopharma; and received consulting fees from Theravance Biopharma. E. Golden and K. Freeman have nothing to disclose. R.K. Bhavaraju-Sanka serves on advisory boards for Amgen and Argenx. H. Snapper has nothing to disclose. G.A. Cook reports research support from Dysautonomia International. Full disclosure form information provided by the authors is available with the full text of this article at [Neurology.org/cp](https://www.neurology.org/cp).

Publication History

Received by *Neurology® Clinical Practice* September 17, 2025. Accepted in final form February 24, 2026. Submitted and externally peer reviewed. The handling editor was Editor Luca Bartolini, MD, FAAN.

References

- Vernino S, Bourne KM, Stiles LE, et al. Postural orthostatic tachycardia syndrome (POTS): state of the science and clinical care from a 2019 National Institutes of Health Expert Consensus Meeting - part 1. *Auton Neurosci*. 2021;235:102828. doi:10.1016/j.autneu.2021.102828
- Freeman R, Wieling W, Axelrod FB, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Auton Neurosci*. 2011;161(1-2):46-48. doi:10.1016/j.autneu.2011.02.004
- Sheldon RS, Grubb BP II, Olshansky B, et al. 2015 Heart rhythm society expert consensus statement on the diagnosis and treatment of postural tachycardia syndrome, inappropriate sinus tachycardia, and vasovagal syncope. *Heart Rhythm*. 2015;12(6):e41-e63. doi:10.1016/j.hrthm.2015.03.029
- Raj SR, Guzman JC, Harvey P, et al. Canadian cardiovascular society position statement on postural orthostatic tachycardia syndrome (POTS) and related disorders of chronic orthostatic intolerance. *Can J Cardiol*. 2020;36(3):357-372. doi:10.1016/j.cjca.2019.12.024
- Benrud-Larson LM, Dewar MS, Sandroni P, Rummans TA, Haythornthwaite JA, Low PA. Quality of life in patients with postural tachycardia syndrome. *Mayo Clinic Proceedings*. 2002;77(6):531-537. doi:10.4065/77.6.531
- Bourne KM, Chew DS, Stiles LE, et al. Postural orthostatic tachycardia syndrome is associated with significant employment and economic loss. *J Internal Med*. 2021;290(1):203-212. doi:10.1111/joim.13245
- Brewster JA, Garland EM, Biaggioni I, et al. Diurnal variability in orthostatic tachycardia: implications for the postural tachycardia syndrome. *Clin Sci (Lond)*. 2012;122(1):25-31. doi:10.1042/CS20110077
- Raj SR, Fedorowski A, Sheldon RS. Diagnosis and management of postural orthostatic tachycardia syndrome. *CMAJ*. 2022;194(10):E378-E385. doi:10.1503/cmaj.211373
- Bryarly M, Phillips LT, Fu Q, Vernino S, Levine BD. Postural orthostatic tachycardia syndrome: JACC focus seminar. *J Am Coll Cardiol*. 2019;73(10):1207-1228. doi:10.1016/j.jacc.2018.11.059
- Goldstein DS, Cheshire WP Jr. The autonomic medical history. *Clin Auton Res*. 2017;27(4):223-233. doi:10.1007/s10286-017-0425-7
- Raj V, Opie M, Arnold AC. Cognitive and psychological issues in postural tachycardia syndrome. *Auton Neurosci*. 2018;215:46-55. doi:10.1016/j.autneu.2018.03.004
- Uppal J, Baker JR, Hira R, et al. Physiological and clinical comparison of active stand and head-up tilt tests in postural orthostatic tachycardia syndrome (POTS). *Auton Neurosci*. 2025;260:103281. doi:10.1016/j.autneu.2025.103281
- Chelimsky G, Chelimsky T. The gastrointestinal symptoms present in patients with postural tachycardia syndrome: a review of the literature and overview of treatment. *Auton Neurosci*. 2018;215:70-77. doi:10.1016/j.autneu.2018.09.003
- Miglis MG, Barwick F. Sleep disorders in patients with postural tachycardia syndrome: a review of the literature and guide for clinicians. *Auton Neurosci*. 2018;215:62-69. doi:10.1016/j.autneu.2018.05.002
- Cook GA Jr., Sandroni P. Management of headache and chronic pain in POTS. *Auton Neurosci*. 2018;215:37-45. doi:10.1016/j.autneu.2018.06.004
- Doherty TA, White AA. Postural orthostatic tachycardia syndrome and the potential role of mast cell activation. *Auton Neurosci*. 2018;215:83-88. doi:10.1016/j.autneu.2018.05.001
- Roma M, Marden CL, De Wandle I, Francomano CA, Rowe PC. Postural tachycardia syndrome and other forms of orthostatic intolerance in Ehlers-Danlos syndrome. *Auton Neurosci*. 2018;215:89-96. doi:10.1016/j.autneu.2018.02.006
- Fu Q, Levine BD. Exercise and non-pharmacological treatment of POTS. *Auton Neurosci*. 2018;215:20-27. doi:10.1016/j.autneu.2018.07.001
- Snapper H, Cheshire WP. Oral and intravenous hydration in the treatment of orthostatic hypotension and postural tachycardia syndrome. *Auton Neurosci*. 2022;238:102951. doi:10.1016/j.autneu.2022.102951
- Garland EM, Gamboa A, Nwazue VC, et al. Effect of high dietary sodium intake in patients with postural tachycardia syndrome. *J Am Coll Cardiol*. 2021;77(17):2174-2184. doi:10.1016/j.jacc.2021.03.005
- Bourne KM, Sheldon RS, Hall J, et al. Compression garment reduces orthostatic tachycardia and symptoms in patients with postural orthostatic tachycardia syndrome. *J Am Coll Cardiol*. 2021;77(3):285-296. doi:10.1016/j.jacc.2020.11.040
- van Dijk N, Quartieri F, Blanc JJ, et al.; PC-Trial Investigators. Effectiveness of physical counterpressure maneuvers in preventing vasovagal syncope: the physical counterpressure manoeuvres trial (PC-Trial). *J Am Coll Cardiol*. 2006;48(8):1652-1657. doi:10.1016/j.jacc.2006.06.059
- Gibbons CH, Silva G, Freeman R. Cardiovascular exercise as a treatment of postural orthostatic tachycardia syndrome: a pragmatic treatment trial. *Heart Rhythm*. 2021;18(8):1361-1368. doi:10.1016/j.hrthm.2021.01.017
- Ziaks L, Johnson K, Schiltz K, et al. Adaptive approaches to exercise rehabilitation for postural tachycardia syndrome and related autonomic disorders. *Arch Rehabil Res Clin Transl*. 2024;6(4):100366. doi:10.1016/j.jarrct.2024.100366
- Wheatley-Guy CM, Shea MG, Parks JK, et al. Semi-supervised exercise training program more effective for individuals with postural orthostatic tachycardia syndrome in randomized controlled trial. *Clin Auton Res*. 2023;33(6):659-672. doi:10.1007/s10286-023-00970-w
- Fu Q, Vangundy TB, Galbreath MM, et al. Cardiac origins of the postural orthostatic tachycardia syndrome. *J Am Coll Cardiol*. 2010;55(25):2858-2868. doi:10.1016/j.jacc.2010.02.043

27. George SA, Bivens TB, Howden EJ, et al. The international POTS registry: evaluating the efficacy of an exercise training intervention in a community setting. *Heart Rhythm*. 2016;13(4):943-950. doi:10.1016/j.hrthm.2015.12.012
28. Miller AJ, Raj SR. Pharmacotherapy for postural tachycardia syndrome. *Auton Neurosci*. 2018;215:28-36. doi:10.1016/j.autneu.2018.04.008
29. Y Lei L, S Chew D, K Sandhu R, S Sheldon R, R Raj S. Non-pharmacological and pharmacological management of cardiac dysautonomia syndromes. *J Atr Fibrillation*. 2020;13(1):2395. doi:10.4022/jafib.2395
30. Sheldon R, Faris P, Tang A, et al. Midodrine for the prevention of vasovagal syncope: a randomized clinical trial. *Ann Intern Med*. 2021;174(10):1349-1356. doi:10.7326/M20-5415
31. Sheldon R, Raj SR, Rose MS, et al. Fludrocortisone for the prevention of vasovagal syncope: a randomized, placebo-controlled trial. *J Am Coll Cardiol*. 2016;68:1-9. doi:10.1016/j.jacc.2016.04.030
32. Chen L, Wang L, Sun J, et al. Midodrine hydrochloride is effective in the treatment of children with postural orthostatic tachycardia syndrome. *Circ J*. 2011;75(4):927-931. doi:10.1253/circj.cj-10-0514
33. Ross AJ, Ocon AJ, Medow MS, Stewart JM. A double-blind placebo-controlled cross-over study of the vascular effects of midodrine in neuropathic compared with hyperadrenergic postural tachycardia syndrome. *Clin Sci (Lond)*. 2014;126(4):289-296. doi:10.1042/CS20130222
34. Raj SR, Levine BD. *Postural Tachycardia Syndrome (POTS) Diagnosis and Treatment: Basics and New Developments*. American College of Cardiology; 2013.
35. Veazie S, Peterson K, Ansari Y, et al. Fludrocortisone for orthostatic hypotension. *Cochrane Database Syst Rev*. 2021;5(5):CD012868. doi:10.1002/14651858.CD012868.pub2
36. Palma JA, Norcliffe-Kaufmann L, Fuente-Mora C, Percival L, Mendoza-Santesteban C, Kaufmann H. Current treatments in familial dysautonomia. *Expert Opin Pharmacother*. 2014;15(18):2653-71. doi:10.1517/14656566.2014.970530
37. Cappato R, Castelvécchio S, Ricci C, et al. Clinical efficacy of ivabradine in patients with inappropriate sinus tachycardia: a prospective, randomized, placebo-controlled, double-blind, crossover evaluation. *JACC*. 2012;60(15):1323-1329. doi:10.1016/j.jacc.2012.06.031
38. Okamoto LE, Urechie V, Rigo S, et al. Hyperadrenergic postural tachycardia syndrome: clinical biomarkers and response to guanfacine. *Hypertension*. 2024;81(11):2237-2247. doi:10.1161/HYPERTENSIONAHA.124.23035

How to cite this article: Robbins NM, Golden EP, Freeman KC, et al. Approach to postural orthostatic tachycardia syndrome. *Neurol Clin Pract*. 2026;16(3):e200620. doi: 10.1212/CPJ.0000000000200620